Optics Assignment

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(1.) The slit experiment gives the lowest wavelength resolution because a single-slit diffraction pattern is broad, with wide minima and maxima. Broad fringes mean small wavelength changes do not shift the pattern enough to detect they have low resolving power.

(2.) A michelson interferometer converts a tiny wavelength change into a shift of many fringes. More counted fringes gives longer effective optical path difference thus much smaller $\Delta\lambda$ can be detected, thus giving higher resolution.

(3.) The sharpness of Fabry-Perot fringes is measured by a parameter called Finesse (F). It tells how many times the light effectively circulates between the mirrors. Higher the reflectivity, more no. of times the light will reflect, thus the fringes become narrower and sharper and thus we can distinguish wavelengths that differ by a very tiny difference.

$$F = \frac{\pi\sqrt{R}}{1-R}, \quad R = \text{Reflectivity of mirror}$$

(4) Fabry-Perot resolves wavelengths too close for Michelson because multiple reflections inside the cavity create very high finesse, producing extremely narrow transmission peaks. This makes tiny differences in wavelength produce large changes in transmitted intensity thus giving superior resolution compared to Michelson.

(5.) $\because \; x = \frac{\lambda L}{d}$

$$\Rightarrow \boxed{\Delta\lambda = \frac{\Delta x d}{L}}$$

Also, $\Delta x = 0.5\,mm$ $\quad \Rightarrow$ so; $\Delta\lambda = \frac{0.5\times10^{-3}\times 0.2\times10^{-3}}{1}$

$d = 0.2\,mm$

& $L = 1\,m$ $\quad = 1.0\times10^{-7}\,m$

$$\boxed{\Delta\lambda = 100\,nm}$$

(6.) $\because$ The mirror moves by $\Delta S = 5\,\mu m$
& observed $N = 3000$ fringes.

$\because \; N = \frac{2\Delta S}{\lambda}$ $\qquad$ (As; $\Delta N = \pm 1$)

$\Rightarrow \lambda = \frac{2\Delta S}{N}$

So; $\frac{\Delta\lambda}{\lambda} = \frac{\Delta N}{N} = \frac{1}{3000} = 3.33\times10^{-4}$

i.e, (0.033%)

(7.) $F = \frac{\pi\sqrt{R}}{1-R}$, $R = 0.9$

$= \frac{\pi\sqrt{0.9}}{1-0.9} = 29.8$

spectral resolving power $\Rightarrow \frac{\lambda}{\Delta\lambda} = mF$ (m = diffraction order)

$\Rightarrow$ So; $\frac{\Delta\lambda}{\lambda} = \frac{1}{mF}$

So; if $m=1$: $\frac{\Delta\lambda}{\lambda} = \frac{1}{29.8} = 3.36\times10^{-2}$

if $m=100$: $\frac{\Delta\lambda}{\lambda} = \frac{1}{100\times29.8} = 3.36\times10^{-4}$

and so on for other value of m.

(8.) An interferometer gets better wavelength resolution when light travels a longer effective path. So, the hack is to make the light bounce or reflect many times, so the optical path becomes much larger than the physical size.

For example:

In Fabry-Perot cavity, two highly reflective mirrors force light to have multiple reflections between them, thus effective path becomes from meters to Km's inside a small cavity. This greatly increases ~~path~~ phase sensitivity.

(9.) The factors that limits maximum usable path difference are:

(a.) Coherence length of the source: If path difference exceeds coherence length of the light source, then fringes disappear or appear faint. Like we can the difference between the fringe appearance of He-Ne laser and Na-lamp.

(b.) Beam overlap & alignment: If beams stop overlapping perfectly, then interference contrast drops.

(c.) Mechanical Stability: Tiny Vibrations or temperature changes cause phase change then also fringes become blurry.

(d.) Dispersion/Absorption due to medium: The medium through with the light is passing may distort or attenuate the beam.

(10) For better resolution than a standard Fabry-Perot we must increase finesse and effective path length:

To increase finesse ⇒ We can use ultra-high reflectivity mirrors with ($R \simeq 0.999$). Also we should put the cavity in Vacuum and stabilize the temperature to reduce losses.

To increase ~~number of~~ effective path length ⇒ We can use a multi-pass cavity such as white cell / Herriott, in which light is bounced 20-500 times between curved mirrors or we can also use ring cavity or recirculating loop is which light circulates tens of thousands of times before leaking out.

⇒ Thus more no. of passes leads to more phase accumulation and thus a tiny change in λ, shifts many fringes thus giving very high resolution.

(11.) $\because \boxed{\frac{\Delta\lambda}{\lambda} \simeq \frac{1}{F.N}}$ & $\frac{\Delta\lambda}{\lambda} = 10^{-10}$

So; $F.N = \frac{1}{10^{-10}} = 10^{10}$

i.e if we can achieve finesse $= 10^{4}$, then we must count $N = 10^{6}$ fringes.

or

if we can count $N = 10^{7}$ fringes, then the needed finesse is 10^{3}.

(12.) The universal recipe for designing precision interferometer are:

→ Using long cavities or multiple reflections, so phase changes accumulate more, giving finer wavelength sensitivity.

→ Using laser with long coherence length, ensuring the fringes remain clear even at large path differences.

→ Using high reflectivity mirrors & low loss cavity leading to narrow & fine fringes thus small $\Delta\lambda$ detectable.

→ Isolation from vibrations using optical bench/breadboard.

(13) **Concept:** I propose a compact interferometer that uses two Fabry-Perot etalons aligned in series, where the 1^{st} etalon is slightly tilted and the 2^{nd} has a slightly different cavity spacing. This tilt introduces angle dependent resonance, while the spacing mismatch creates a Vernier effect, greatly increasing ~~A~~ wavelength sensitivity without needing a long delay arm like a Michelson.

Optical Design:

→ Collimated input beam
→ Etalon A (tilted by a small angle $\theta \sim 1^\circ$) ⇒ it generates spatially shifted interference fringes.

→ Etalon B (with spacing $L_2 = L_1(1+\epsilon)$, where $\epsilon <<< 1$)
↳ produces Vernier amplification.
→ Lens focusing the transmitted pattern onto a camera or photodiode array.

⇒ **Physical Principle used:**

⇒ **Fabry - Perot resonance:**

Each etalon transmits wavelengths that satisfy $2nL = m\lambda$.

⇒ Tilt - induced dispersion : ~~A~~ Tilting etalon A creates a slight change in effective optical thickness across the beam, thus spatially varying resonance peaks.

⇒ Vernier Amplification: since etalon B's spacing differs by a small fraction ε, the combined transmission only peaks when both etalons align in resonance, giving an effective line width much narrower than either etalon alone.

Wavelength resolution Estimate:

For a single etalon,

$$\delta\lambda_{FP} \approx \frac{\lambda^2}{2nLF}$$

eg:

$\lambda = 1550nm$

$L = 10mm$

Finesse, $F = 300$

then; $\delta\lambda_{FP} = 4 \times 10^{-4}\ nm$

with $\varepsilon = 0.001$;

$$\delta\lambda_{Vernier} \approx \delta\lambda_{FP}\,\varepsilon \approx 4 \times 10^{-7}\ nm$$

So, the theoretical wavelength resolution is of the order of

$\boxed{4 \times 10^{-7}\ nm}$